



Optimal FCP Production Level for *PRT BELL*

Team #381

Kashyap Mehta

Ananya Jain

Ning Wei

Ji-Lun Mo



Background

PRT BELL specializes in the production of food colors, an essential ingredient in the food processing industry. The company has established itself as a key player in this niche market.

The company offers a diverse range of food colors, catering to various industry needs. Their products might include natural, synthetic, and organic color options, ensuring compatibility with a wide array of food products.

They might face challenges such as fluctuating raw material costs, regulatory changes, and stiff competition. However, opportunities like increasing demand for natural and organic food colors, and expansion into emerging markets, provide growth prospects.



Problem Statement

For the problem at hand, we aim to optimize the number of units produced and sold, with an initial budget of \$100,000. Moreover, the first year profit can be carry into second year as budget. The company uses advertising to create demand so per dollar invested in advertising, we have a forecasted demand of x units which is variable for different finished goods and year. To maintain a diverse portfolio of products, the company also has constraints of sales ratio between each product.

	YEAR 1				YEAR 2			
Products	Red	Orange	Yellow	Green	Red	Orange	Yellow	Green
Selling price per unit	50	60	55	45	55	45	60	40
Demand per ad spent	8	10	7	8	10	12	9	9



This case involving PRT BELL focuses on finding the optimal production level of four Food Colour Preparations to maximize profit. This entails a careful analysis of production costs, market demand, and pricing strategies. By optimizing production, PRT BELL aims to efficiently meet market demand while maximizing its profitability.

Variable Cost of production	Per Unit
Labor Cost	4
Packaging Cost	2

Sales Ratio	MIN	MAX
FCP RED	20%	40%
FCP ORANGE	20%	40%
FCP YELLOW	10%	30%
FCP GREEN	5%	15%



Ratio	FCP Red	FCP Orange	FCP Yellow	FCP Green	Cost of RM
Base	90%	90%	90%	90%	\$3
Dye 1	5%	0	10%	0	\$7
Dye 2	5%	5%	0	0	\$8
Dye 3	0	5%	0	0	\$7
Dye 4	0	0	0	10%	\$9



Model

Inputs:

- $i = 1, 2, 3, 4, 5$; index to represent raw materials (e.g. 1= Dye 1, ..., 5= base)
- $j = 1, 2, 3, 4$; index to represent finished goods (Red, Orange, Yellow, Green)
- $k = 1, 2$; index of year
- C_i = Cost of raw materials i per kg [e.g. $C_1 = \$7$]
- P_{jk} = Price of finished good j per unit in year k [e.g. $P_{11} = \$50$]
- R_{ij} = Percentage of raw material i required to produce finished good j [e.g. $R_{11}=5$]
- D_{jk} = Demand created (boxes) per dollar of advertising on good j in year k
[e.g. $D_{11} = 1$]
- LC = Labor Cost per unit of finished good [e.g. $LC = \$4$]
- PC = Packaging Cost per unit of finished good [e.g. $PC = \$2$]
- IB = Initial budget in year 1 [e.g. $IB = \$100,000$]
- Min_j = The minimum percentage of each finished good j [e.g. $Min_1 = 20\%$]
- Max_j = The maximum percentage of each finished good j [e.g. $Max_1 = 40\%$]



Decision Variables

X_{jk} : Number of finished good j to produce in year k

Y_{jk} : Number of finished good j to sell in year k

Z_{jk} : Cost of advertisements on finished good j in year k

Calculate Parameters

RM_{ijk} : Raw material i used in j good in year k [$(X_{jk} * 5) * R_{ij}$]

TC_k : Total cost in year k [$\sum RM_{ijk} + (\sum PC * X_{jk}) + (\sum LC * X_{jk})$]



Objective and Objective Function

To Maximize the Profit

$$\max [\sum_k (\sum_j Y_{jk} * P_{jk}) - TC_k]$$

Constraints



$X_{jk} \geq 0$;integer

$Y_{jk} \geq 0$;integer

$Z_{jk} \geq 0$;integer

$[Y_{j1}] \leq [X_{j1}]$

[In year 1, the available goods is the produced good, and the number of sold goods in year 1 have to be smaller than the available goods]

$[Y_{j2}] \leq [X_{j2} + X_{j1} - Y_{j1}]$

[In year 2, the available goods is the produced goods plus the goods that are not sold in year 1, and the number of sold goods in year 2 has to be smaller than it]

$[Y_{jk}] \leq [Z_{jk} * D_{jk}]$

[The number of sold goods has to be smaller than the demand we got from advertisement]

$TC_1 \leq IB$

[The cost of year 1 has to be smaller than the initial budget]

$TC_2 \leq IB - TC_1 + (Y_{jk} * P_{jk})$ [The cost of year 2 has to be smaller than the profit from year 1 plus the initial budget]

$\min j \leq Y_{jk} / (\sum_j Y_{jk}) \leq \max j$ [The percentage of sold goods has to be in the range]

SOLVER

PRT BELL

Raw Material

Base	\$3.00
Dye 1	\$7.00
Dye 2	\$8.00
Dye 3	\$7.00
Dye 4	\$9.00

Selling prices

	Year 1	Year 2
FCP RED	\$50.00	\$55.00
FCP ORANGE	\$60.00	\$45.00
FCP YELLOW	\$55.00	\$60.00
FCP GREEN	\$45.00	\$40.00

Demand created (boxes) per dollar of advertising

	Year 1	Year 2
FCP RED	8	10
FCP ORANGE	10	12
FCP YELLOW	7	9
FCP GREEN	8	9

Packaging Cost

2

Labour Cost

4

RAW MATERIAL USED - KG	TOTAL	Base	Dye 1	Dye 2	Dye 3	Dye 4
FCP RED	5,365.00	4,828.50	268.25	268.25	-	-
FCP ORANGE	8,545.00	7,690.50	-	427.25	427.25	-
FCP YELLOW	6,420.00	5,778.00	642.00	-	-	-
FCP GREEN	1,095.00	985.50	-	-	-	109.50

RAW MATERIAL USED - KG	TOTAL	Base	Dye 1	Dye 2	Dye 3	Dye 4
FCP RED	20,290.00	18,261.00	1,014.50	1,014.50	-	-
FCP ORANGE	12,635.00	11,371.50	-	631.75	631.75	-
FCP YELLOW	15,235.00	13,711.50	1,523.50	-	-	-
FCP GREEN	2,600.00	2,340.00	-	-	-	260.00

Year 1		Material Cost Packaging		Labour	Year 2	Material Cost Packaging		Labour	Ratio	FCP Red	FCP Orange	FCP Yellow	FCP Green		
FCP RED		18,509.25	2,146.00	4,292.00	FCP RED	70,000.50	8,116.00	16,232.00	Base	90%	90%	90%	90%		
FCP ORANGE		29,480.25	3,418.00	6,836.00	FCP ORANGE	43,590.75	5,054.00	10,108.00	Dye 1	5%	0	10%	0		
FCP YELLOW		21,828.00	2,568.00	5,136.00	FCP YELLOW	51,799.00	6,094.00	12,188.00	Dye 2	5%	5%	0	0		
FCP GREEN		3,942.00	438.00	876.00	FCP GREEN	9,360.00	1,040.00	2,080.00	Dye 3	0	5%	0	0		
									Dye 4	0	0	0	10%		
Dollars on advertising															
		Year 1	Year 2			Year 1	Year 2	Cash constraints							
Produced	FCP RED	1,073	4,058	FCP RED		\$134	\$406	Spent	\$	99,985	\$	236,675			
	FCP ORANGE	1,709	2,527	FCP ORANGE		\$171	\$211			<=		<=			
	FCP YELLOW	1,284	3,047	FCP YELLOW		\$183	\$338								
	FCP GREEN	219	520	FCP GREEN		\$27	\$58	Available	\$	100,000	\$	236,680			
									Revenue	\$	236,665	\$	540,405		
	FCP RED	1,073	4,058					Profit							
	FCP ORANGE	1,709	2,527							\$	440,410				
	FCP YELLOW	1,284	3,047												
	FCP GREEN	219	520												
		<=	<=			Min %	Year 1 min	Year 2 min	Actual Year 1		Actual Year 2		Max %	Year 1 max	Year 2 max
Sold	FCP RED	1,073	4,058	FCP RED		20%	857	2,030	<=	1,073	4,058	<=	40%	1,714	4,060
	FCP ORANGE	1,709	2,527	FCP ORANGE		20%	857	2,030	<=	1,709	2,527	<=	40%	1,714	4,060
	FCP YELLOW	1,284	3,045	FCP YELLOW		10%	429	1,015	<=	1,284	3,045	<=	30%	1,286	3,045
	FCP GREEN	219	520	FCP GREEN		5%	214	508	<=	219	520	<=	15%	643	1,523
			<=	<=											
Demand	FCP RED	1,073	4,058												
	FCP ORANGE	1,709	2,527												
	FCP YELLOW	1,284	3,045												
	FCP GREEN	219	520												



Conclusion

The model has suggested a solution that will generate a two years total profit of \$440,410. The setup explains the dollar amount to spent on advertising to create a reasonable demand as shown below.

Dollars on advertising

	Year 1	Year 2
FCP RED	\$134	\$406
FCP ORANGE	\$171	\$211
FCP YELLOW	\$183	\$338
FCP GREEN	\$27	\$58

Conclusion

In the next table, we can find some produced goods are not sold. The optimal solution taken reality market into consideration so there might be surplus for certain products.

Cash constraints

		Year 1	Year 2
Spent	\$	99,985	\$ 2,36,675
		<=	<=
Available	\$	1,00,000	\$ 2,36,680
Revenue	\$	2,36,665	\$ 5,40,405
Profit	\$	4,40,410	

		Year 1	Year 2
Produced	FCP RED	1,073	4,058
	FCP ORANGE	1,709	2,527
	FCP YELLOW	1,284	3,047
	FCP GREEN	219	520
Available	FCP RED	1,073	4,058
	FCP ORANGE	1,709	2,527
	FCP YELLOW	1,284	3,047
	FCP GREEN	219	520
Sold		<=	<=
	FCP RED	1,073	4,058
	FCP ORANGE	1,709	2,527
	FCP YELLOW	1,284	3,045
Demand	FCP GREEN	219	520
		<=	<=
	FCP RED	1,073	4,058
	FCP ORANGE	1,709	2,527
	FCP YELLOW	1,284	3,045
	FCP GREEN	219	520

Overall, we recommend PRT BELL to follow the optimal capital allocation strategy, which provides a guideline for optimal production level while maximizing the profit under limited budget.



Improvement Solutions

- Include available labor hours and labor costs
- Integrate raw materials costs fluctuations
- Consider different product machines and their capacities
- Account for wastage in productions
- Forecasted demand based on seasonal trends



THANK YOU!